Name- Bhakare Purva Shivaji

Roll no- 08

Batch-T1

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
# 1. Load Diabetes datase
file_path = r"C:\Users\PVGIT\Downloads\archive (7).zip"
df = pd.read_csv(file_path)
print("First 5 rows:")
print(df.head())
# 2. Data Cleaning
print("\nDataset Information:")
df.info()
# 3. Handle Missing Values
df["Glucose"] = df["Glucose"].fillna(df["Glucose"].mean())
df["BloodPressure"] = df["BloodPressure"].fillna(df["BloodPressure"].mean())
df["SkinThickness"] = df["SkinThickness"].fillna(df["SkinThickness"].mean())
df["Insulin"] = df["Insulin"].fillna(df["Insulin"].mean())
df["BMI"] = df["BMI"].fillna(df["BMI"].mean())
print("\nMissing values:")
print(df.isnull().sum())
# 4. Encoding
# Diabetes dataset already contains numerical values
# 5. Statistical Analysis
print("\nStatistical Analysis:")
print(df.describe())
# 6. Select Input and Output
X = df[[
```

```
    "Pregnancies",
    "Glucose",
    "BloodPressure",
    "SkinThickness",
    "Insulin",
    "BMI",
    "DiabetesPedigreeFunction",
    "Age"
]]
y = df["Outcome"]
# 7. Split Dataset
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)
print("\nTraining data:", X_train.shape)
print("Testing data:", X_test.shape)
# 8. Feature Scaling
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
print("\nScaled Training Data:")
print(X_train[:5])
# 9. Histogram
df.hist(figsize=(10, 8))
plt.suptitle("Histograms")
plt.tight_layout()
plt.show()
# 10. Boxplot
```

```
plt.figure(figsize=(8, 5))
df[[
    "Glucose",
    "BloodPressure",
    "SkinThickness",
    "Insulin",
    "BMI",
    "Age"
]].boxplot()
plt.title("Boxplot")
plt.ylabel("Values")
plt.show()
```

OUTPUT-

First 5 rows:

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunction	Age	Outcome
0	6	148	72	35	0	33.6	0.627	50	1
1	1	85	66	29	0	26.6	0.351	31	0
2	8	183	64	0	0	23.3	0.672	32	1
3	1	89	66	23	94	28.1	0.167	21	0
4	0	137	40	35	168	43.1	2.288	33	1

Dataset Information:

<class 'pandas.DataFrame'>

RangeIndex: 768 entries, 0 to 767

Data columns (total 9 columns):

#	Column	Non-Null Count	Dtype
0	Pregnancies	768 non-null	int64
1	Glucose	768 non-null	int64
2	BloodPressure	768 non-null	int64
3	SkinThickness	768 non-null	int64
4	Insulin	768 non-null	int64
5	BMI	768 non-null	float64
6	DiabetesPedigreeFunction	768 non-null	float64
7	Age	768 non-null	int64
8	Outcome	768 non-null	int64

dtypes: float64(2), int64(7)

memory usage: 54.1 KB

Missing values:

Pregnancies	0
Glucose	0
BloodPressure	0
SkinThickness	0
Insulin	0
BMI	0
DiabetesPedigreeFunction	0
Age	0
Outcome	0

dtype: int64

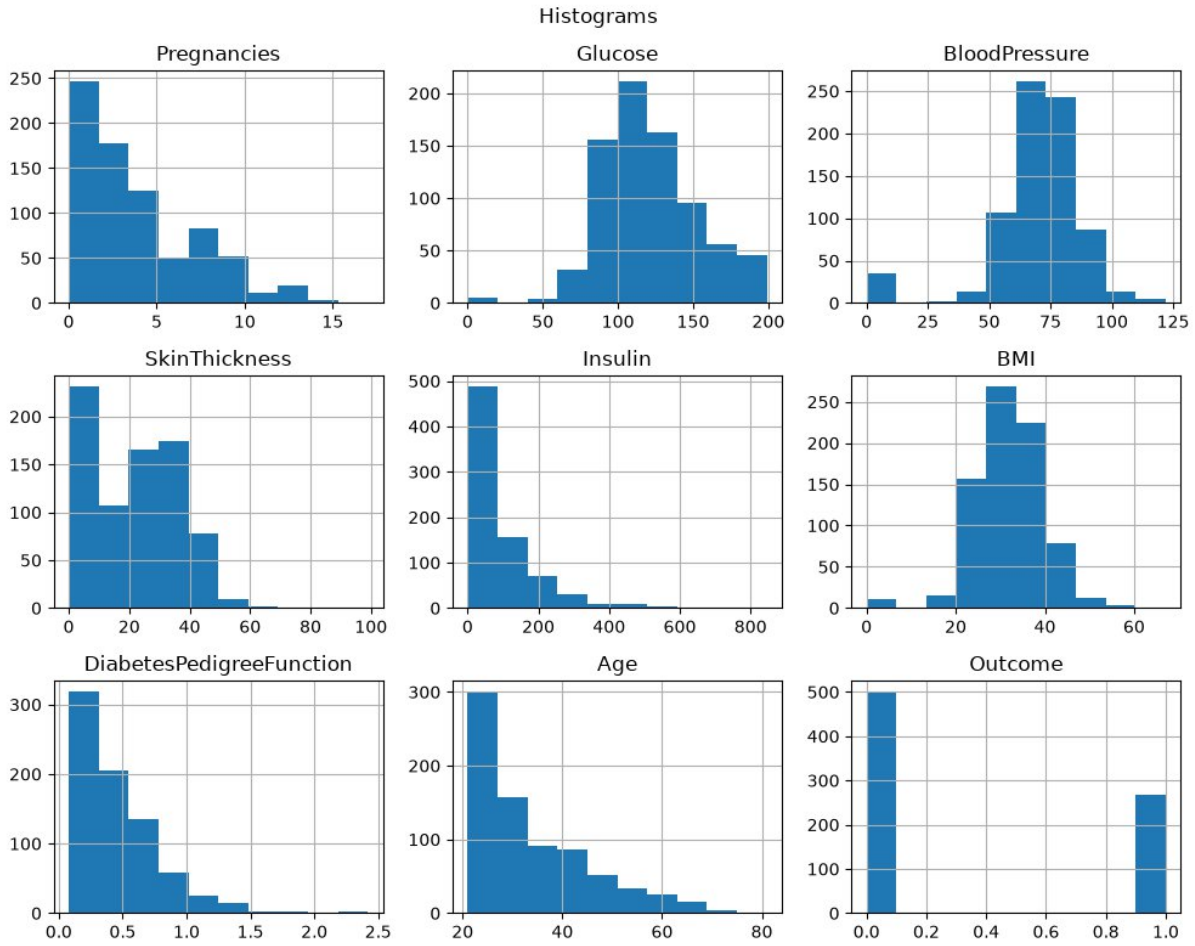
Statistical Analysis:

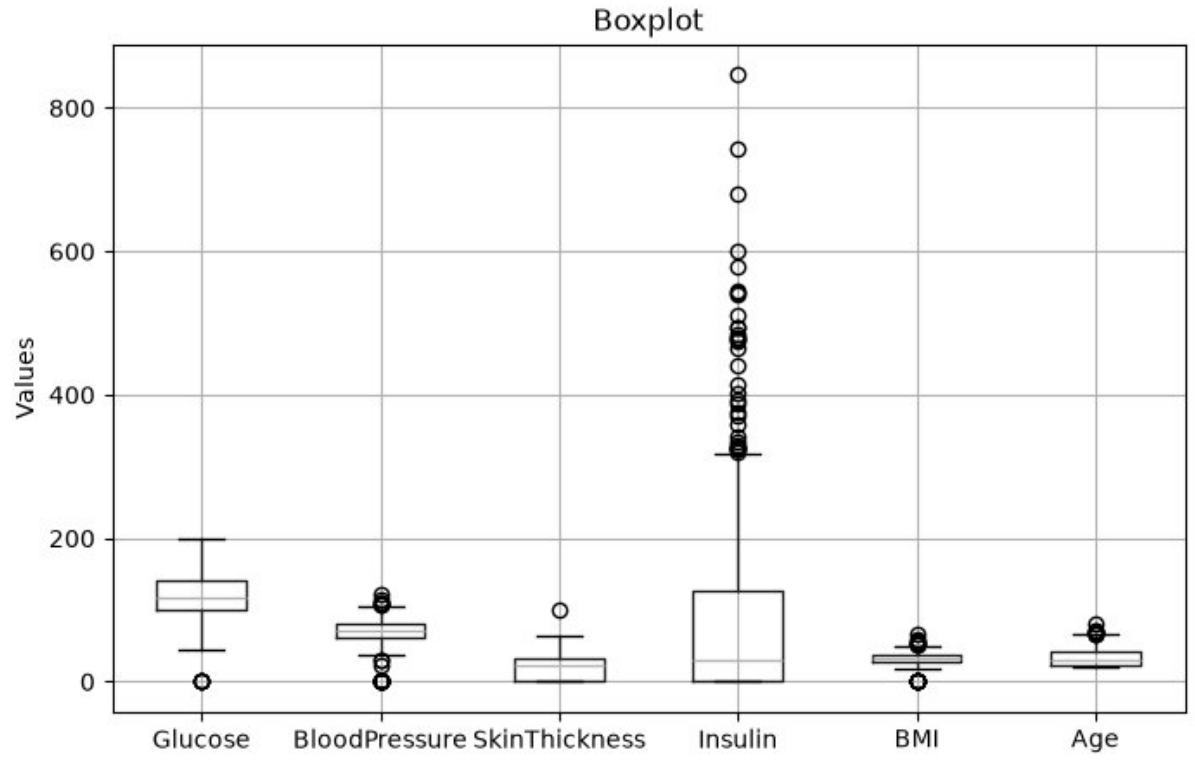
	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin \
count	768.000000	768.000000	768.000000	768.000000	768.000000
mean	3.845052	120.894531	69.105469	20.536458	79.799479

```
std       3.369578   31.972618      19.355807      15.952218  115.244002
min       0.000000    0.000000       0.000000       0.000000    0.000000
25%       1.000000   99.000000      62.000000       0.000000    0.000000
50%       3.000000  117.000000      72.000000      23.000000   30.500000
75%       6.000000  140.250000      80.000000      32.000000  127.250000
max      17.000000  199.000000     122.000000      99.000000  846.000000

              BMI  DiabetesPedigreeFunction        Age     Outcome
count  768.000000              768.000000  768.000000  768.000000
mean    31.992578                0.471876   33.240885    0.348958
std      7.884160                0.331329   11.760232    0.476951
min      0.000000                0.078000   21.000000    0.000000
25%     27.300000                0.243750   24.000000    0.000000
50%     32.000000                0.372500   29.000000    0.000000
75%     36.600000                0.626250   41.000000    1.000000
max     67.100000                2.420000   81.000000    1.000000

Training data: (614, 8)
Testing data: (154, 8)
Scaled Training Data:
[[-0.52639686 -1.15139792 -3.75268255 -1.32277365 -0.70120553 -4.13525578
  -0.49073479 -1.03594038]
 [ 1.58804586 -0.27664283  0.68034485  0.23350519 -0.70120553 -0.48916881
   2.41502991  1.48710085]
 [-0.82846011  0.56687102 -1.2658623  -0.09071957  0.01344832 -0.42452187
   0.54916055 -0.94893896]
 [-1.13052335  1.2541786  -1.04961706 -1.32277365 -0.70120553 -1.30372015
  -0.63929127  2.79212217]
 [ 0.68185612  0.41066475  0.57222224  1.07648956  2.48460077  1.83812075
  -0.68682934  1.13909516]]
```


Histograms
Pregnancies
Glucose
BloodPressure
SkinThickness
Insulin
BMI
DiabetesPedigreeFunction
Age
Outcome


Boxplot
Values
Glucose
BloodPressure
SkinThickness
Insulin
BMI
Age